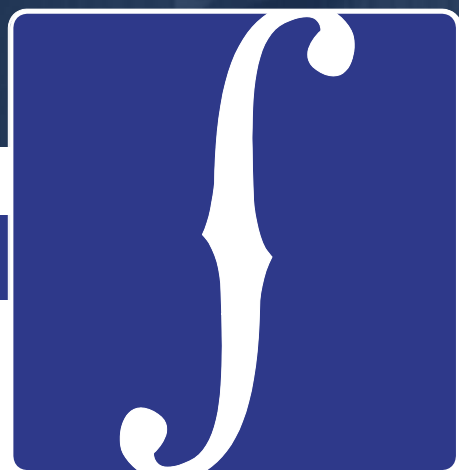




CELLO

Agency Prepayment: Turnover
April 2016

1 Base Turnover – at least 30 bps OTM

- In order to get turnover from total prepayments, we need to subtract Refinances, Defaults and Curtailments.
- To estimate base turnover, we look at speeds on pools that are at least **30 bps OTM** and that consequently should have zero refinancing rates. We then subtract defaults and curtailments to build our base turnover.
- However, the OTM data includes some OTM **Cashout refinances**. We also need to subtract this contribution from our base turnover.

2 Pool Level Data Set

- Consistent with the Curtailment and Default submodels, we use pool level information to fit and track the Turnover submodel.

3 Refinance Incentive

- Incentive acts as a Lock-in factor in the Turnover model. In order to get a more accurate lock-in measurement, we made the following improvements for the pool level incentive calculation:
 - Modified **origination PMI** for HARPed loans (CQ/CR/MHA)
 - Used a loan-level estimate for the average incentive experienced by all loans in a pool instead of relying on the pool WAC.

Turnover model V1

2 Step Fitting

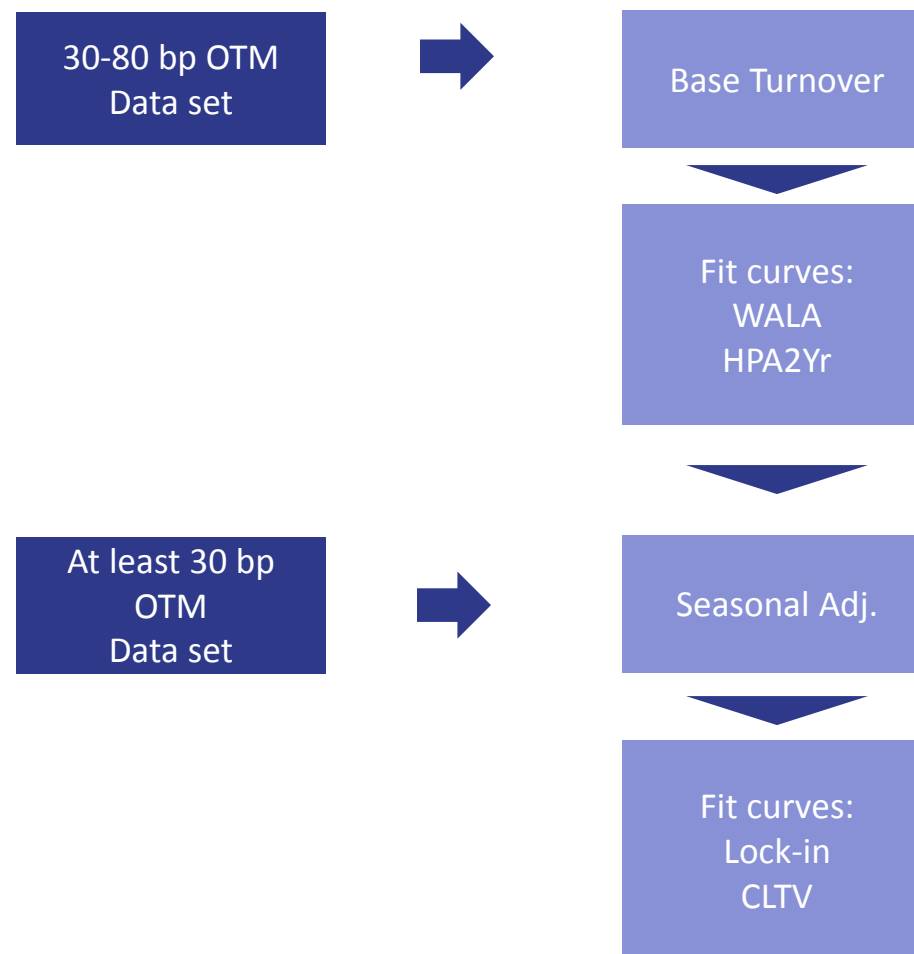
Using two data sets for different incentive ranges

5 Parameters

Seasonal Adjustment
WALA
HPA2Yr
Lock-In
CLTV

1 Adjustment Function

Tweaks for HPA2Yr



Turnover model V1

2 Step Fitting

Using two data sets for different incentive ranges

5 Parameters

Seasonal Adjustment
WALA
HPA2Yr
Lock-In
CLTV

1 Adjustment Functions

Tweaks for HPA2Yr



Updates for v2

1. Subtract OTM Cashout Refinances

The OTM prepayment data includes some OTM cashout refinances. We need to remove that from our base turnover value.

2. Add 1 Additional Parameter

Add **Loan Purpose** as an explanatory variable. The lock-in incentive is now performed at the loan level.

3. Remove 2-step fitting process

Use all the pool level data that is at least 30 bps OTM to do the fitting process in one step.

4. Remove adjustment functions

The old model involved tweaks because we used loan level data to calibrate the model and then used these adjustment factors to fit pool level data

5. Add constraints for extrapolation

Use cap and floors for out of range extrapolation. Make sure the tail performance is reasonable.

Question

- Our previous calculation for HARPed loan incentive was misleading since HARPed loans typically pay no MI or carry over their MI from the previous loan.
- So, what MI was the HARPed borrower paying before they refinanced?

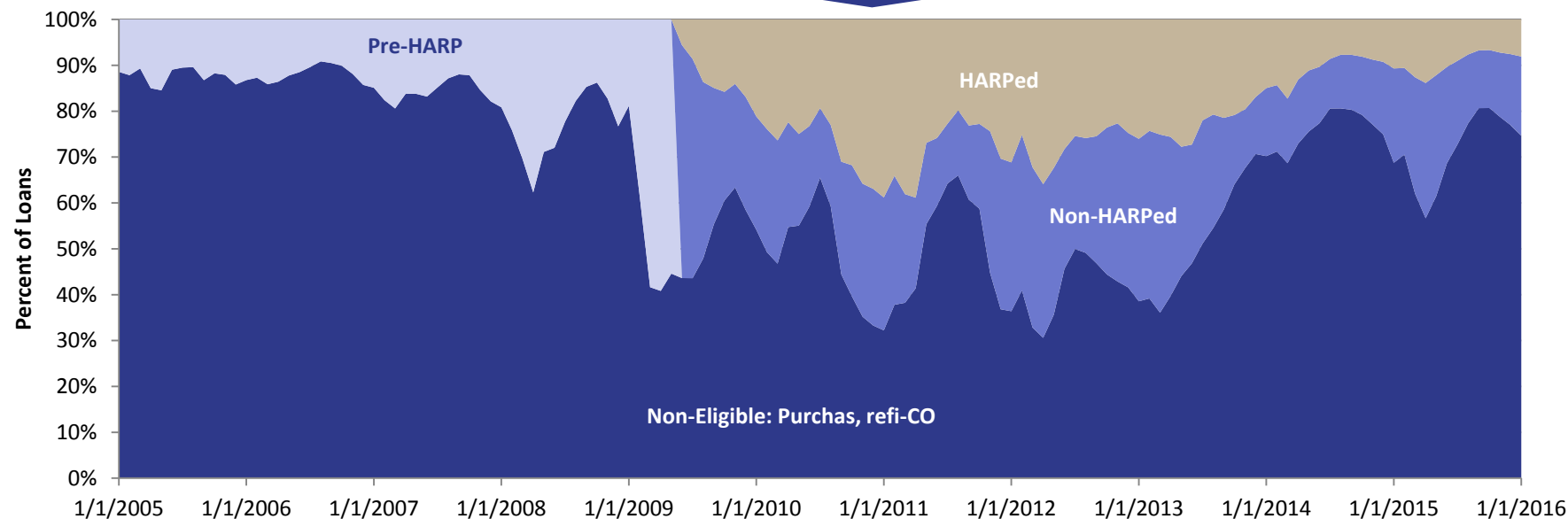
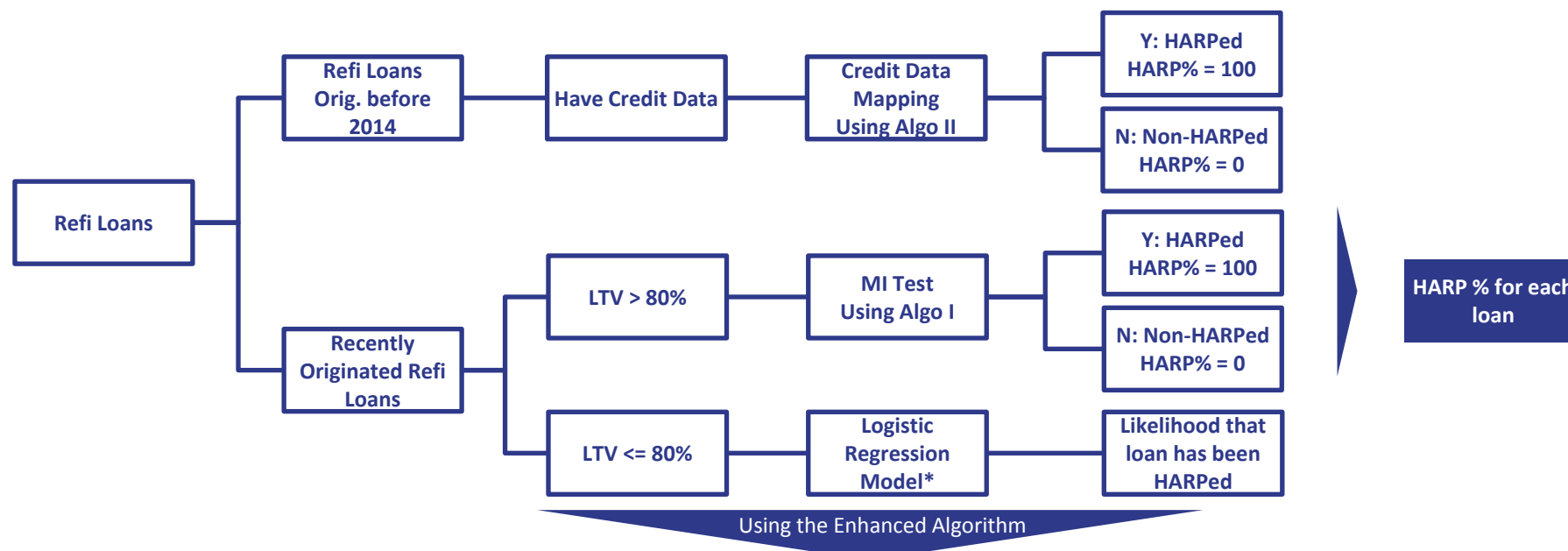
Problem

- We don't have information about the borrower's previous loan. Consequently, there is no straightforward way to determine the PMI at origination.

Solution

- Using our Enhanced HARPed Identification algorithm, we can identify a HARP% for each loan.
- We use the MI coverage level to look up MI rates from before 2009/06 (using MGIC data) for all HARPed loans.

2. Incentive – Improving Identification of HARPed Loans



Based on an enhanced identification algorithm for HARPed loans, we are able to calculate pool PMIs more accurately (at the loan level) which in turn improves the accuracy of the incentive calculation.

Example Loan:

Loan Seq Num: U60490000208

firstPaymtDt	4/1/2011
cs	781
origLTV	121
loanPurposeType	RE-FI-NCO
occType	OWNER
originalLoanAmount	89,320
pctMtglIns	20

As the LTV is greater than the largest number in the table, we assigned the maximum number (or a number of 1000 bps) in the table to the base MI (i.e., we ignore what the credit score is).



We will have a PMI of **110 bps** for the loan.

Find the MI premium based on loan info

BPMI Fixed Monthly Premiums
30-YEAR ■ NONREFUNDABLE

BASE LTV	COVERAGE	Class Selection		
		CLASS 1 720+	CLASS 2 680-719	CLASS 3 < 680
97% to 95.01%	35%	.88%	1.10%	N/A
	30	.80	1.00	N/A
	25	.71	.87	N/A
	18	.59	.77	N/A
95% to 90.01%	35	.76	1.06	1.37%
	30	.67	.94	1.20
	25	.62	.84	1.08
	18	.56	.74	.85
90% to 85.01%	16	.54	.72	.79
	30	.59	.69	.88
	25	.49	.62	.76
	17	.43	.49	.61
85% & UNDER	12	.39	.44	.52
	25	.44	.48	.69
	17	.37	.43	.58
	12	.32	.38	.44
	6	.30	.34	.38

	CLASS 1 720+	CLASS 2 680-719	CLASS 3 < 680
PREMIUM ADJUSTMENTS			
Rate-and-Term Refinance	.10	.15	.30
Second Homes	.14	.20	.35
Loan Amounts > \$417,000	.25	.40	.60
Refundable Monthly Premium	.01	.02	.03
Declining Renewals	.03	.04	.05
Employee Relocation Loans	-.04	-.07	-.10
≤ 25-Year Amortization Term	-.05	-.08	-.11

NOTE: Rates cannot be reduced below .15%.



Example Loan:

Loan Seq Num: U60490000208

firstPaymtDt	4/1/2011
CS	781
origLTV	121
loanPurposeType	RE-FI-NCO
occType	OWNER
originalLoanAmount	89,320
pctMtglIns	20

Based on the historical MI documents, we assign 79 bps MI to the example loan instead of 110 bps.



Makes the incentive 31 bps lower.

Look up the MI coverage

MONTHLY PREMIUMS & ZOMP!			FIXED LEVEL PAYMENTS FOR THE FIRST 5 YEARS	NONFIXED ANNUAL CAPS ≤ 1%	NONFIXED ANNUAL CAPS > 1%
			30-Year	30-Year	30-Year
			1st Year & Renewals	1st Year & Renewals	1st Year & Renewals
			No Refund	No Refund	No Refund
Base LTV (%)	MGIC Coverage (%)	Reduces Exposure To (%)			
95* -90.01	35	62	1.06%	1.21%	1.25%
	30	67	.94	1.04	1.08
	27	70	.88	.94	.98
	25	72	.84	.90	.94
	22	75	.80	.86	.90
	18	78	.74	.79	.82
90 -85.01	16	80	.72	.77	.79
	35	59	.75	.92	.96
	30	63	.69	.84	.89
	25	68	.62	.73	.78
	22	71	.57	.67	.71
	20	72	.52	.59	.64
85 & Under	18	74	.50	.57	.62
	17	75	.49	.56	.61
	12	80	.44	.45	.50
	35	56	.65	.77	.83
	30	60	.57	.67	.71
	25	64	.48	.55	.59
	22	67	.46	.53	.58
	20	68	.45	.51	.55
	18	70	.44	.46	.51
	17	71	.43	.44	.49
	12	75	.38	.39	.44
	6	80	.34	.36	.40

Adjustments shown in the Premium Adjustments for Monthly Premiums & ZOMP! table apply.

Premium Adjustments for Monthly Premiums & ZOMP!

Rate-and-Term Refinance	+ .10%
Second Homes	+ .14%
Loan Amounts > \$417,000	+ .25%
Level Annual Premium (Refundable)	- .04%
Refundable Monthly Premium	+ .01%
25-Year & Under	- .11%
Employee Relocation Loans	
• LTVs greater than 90%	- .12%
• LTVs of 85.01%-90%	- .10%
• LTVs of 85% or less	- .07%
Note: Rates cannot be reduced below .15%.	

NATIONAL RATE CARD
MARCH 2009

Question

- Our current calculation for pool level incentive treats the pool as a single loan and uses the weighted average of the pool's various characteristics (FICO, CLTV etc) to infer PMI and LLPA levels.
- How accurate is the calculation?

Problem

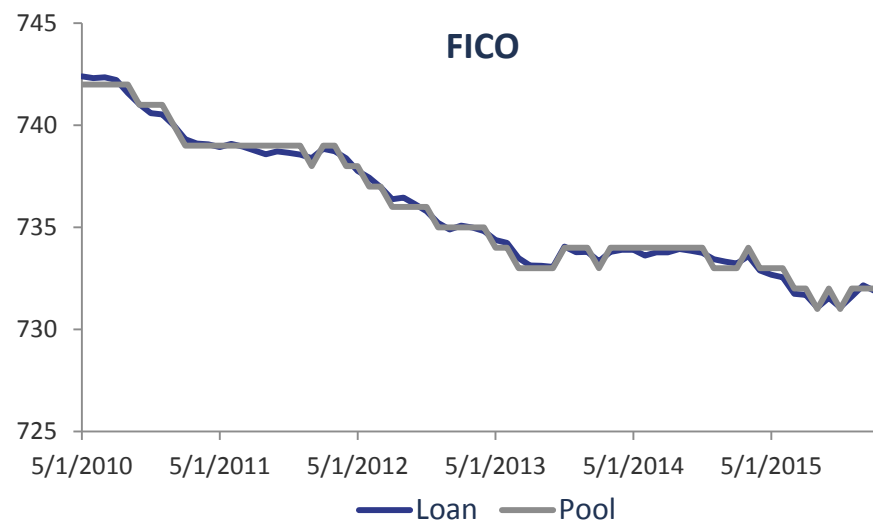
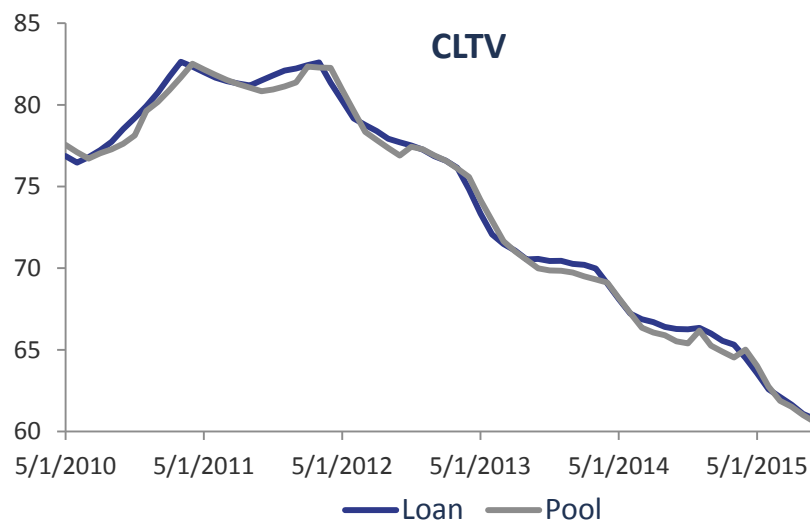
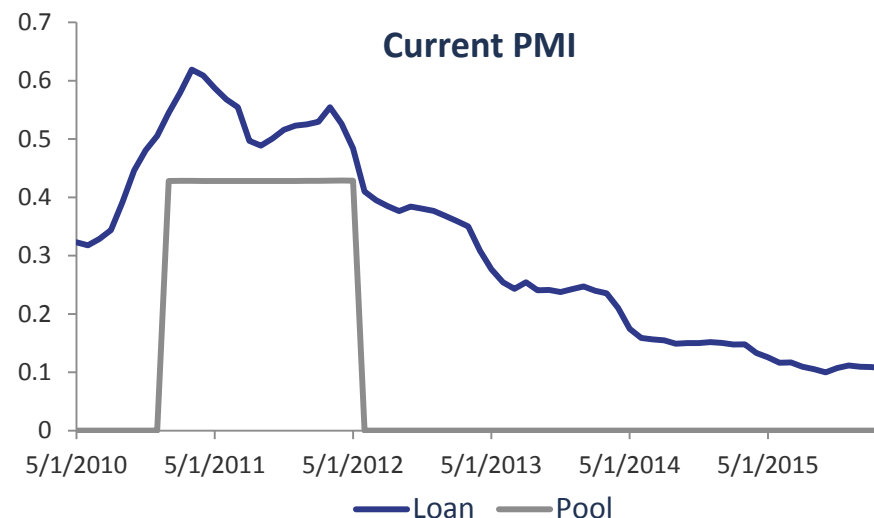
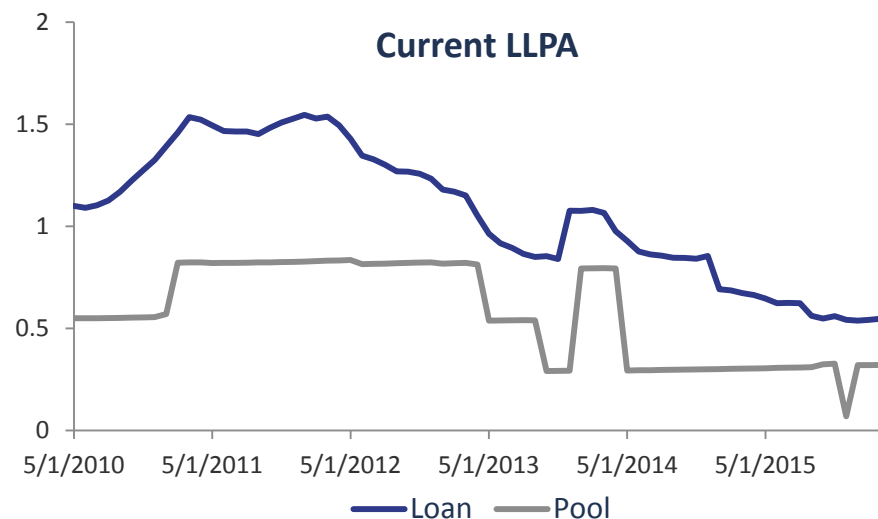
- Pool-level calculations mask dispersion.
- The LLPA and PMI mappings are step functions, not linear functions. So, pool-level calculations that are near step thresholds can have significant biases.

Solution

- Use loan level incentive calculations, including LLPA and PMI.
- Refine calculation of loan PMI for HARPed loans
- Aggregate the loan level incentives to estimate pool incentive.

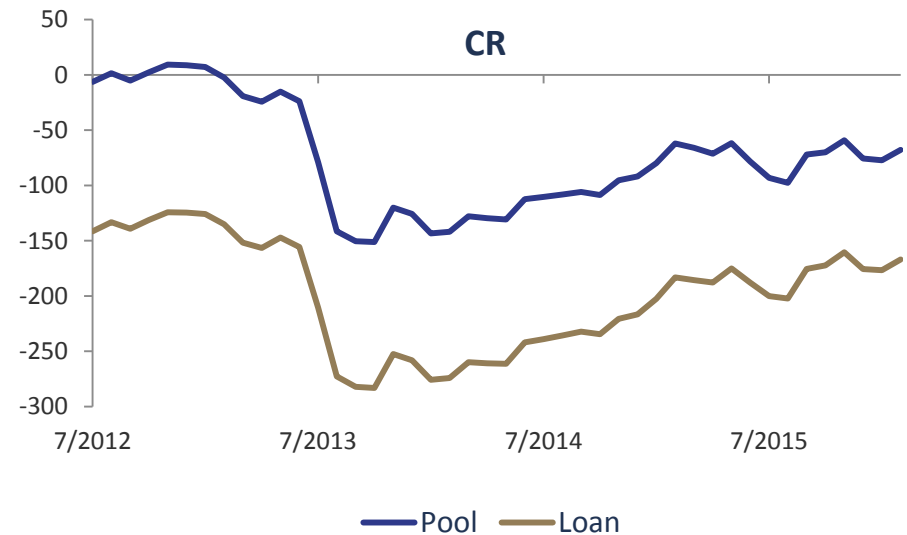
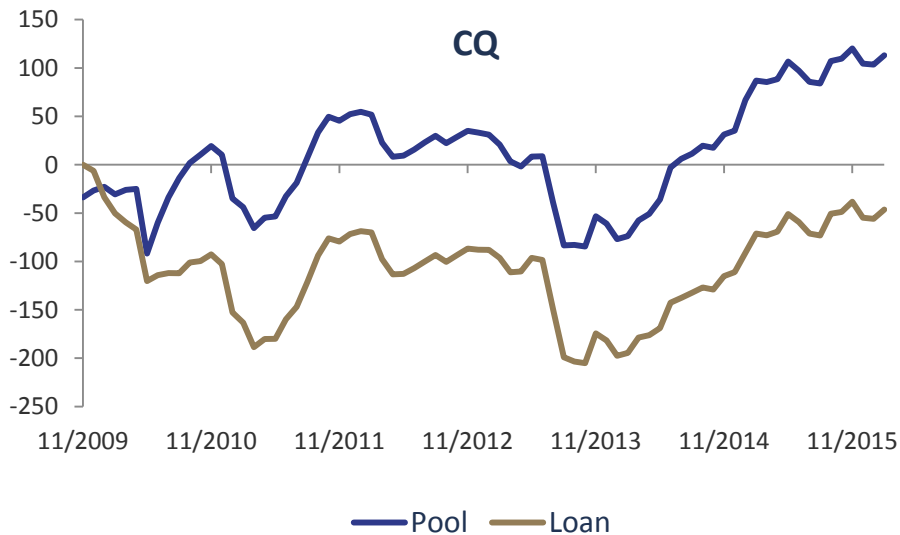
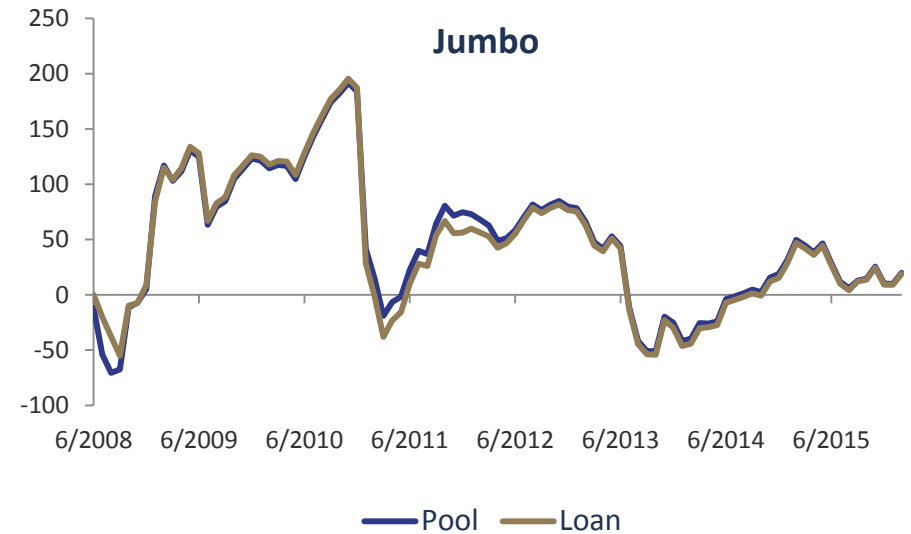
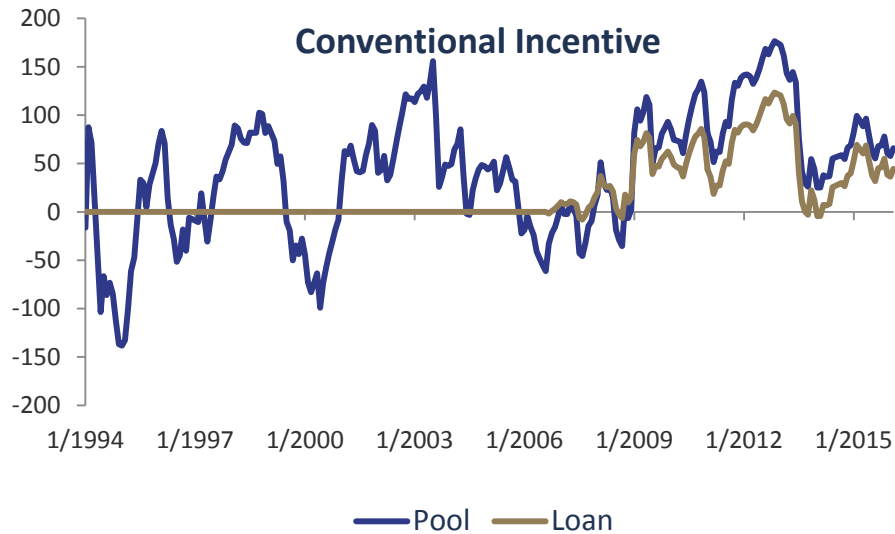
- The Impact of Loan Dispersion for calculating various pool characteristics**

There is little difference between the pool- and loan-level calculations for FICO and CLTV. For LLPA and PMI calculations, the differences can be substantial because of the step-function effect.



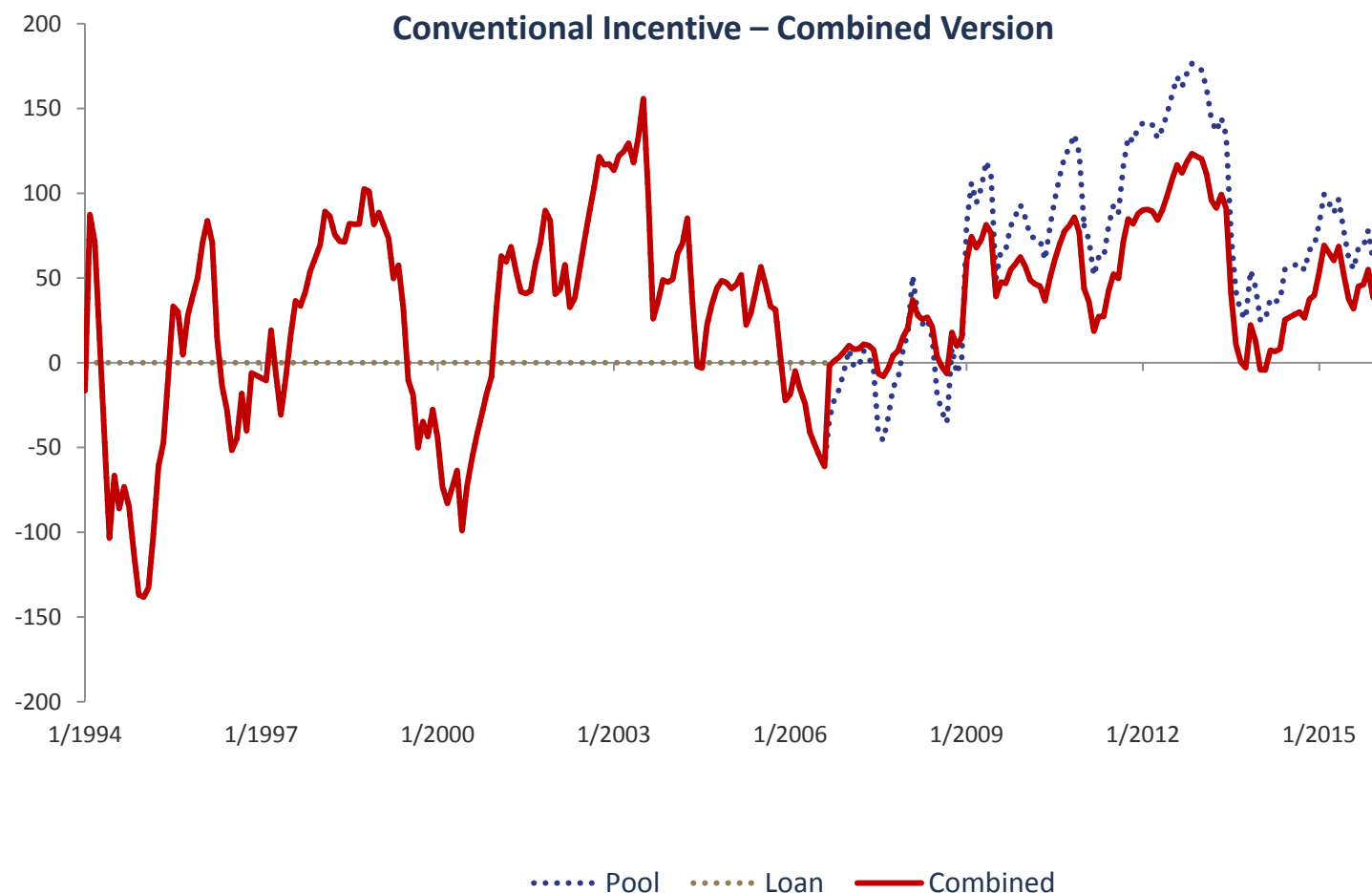
7. Incentive – Loan-level vs Pool-level for Different Tickers

- **Loan- vs Pool-level aggregations for Incentive** (inclusive of fixing the Origination PMI for HARPed Loans):



- Combined Version for Conventional Pools**

To handle the problem that there is no loan level data before 2006, we use pool level data before 2006. The combined version is relatively reliable because the impact of LLPA/PMI is minor prior to 2006.



Question

- 30bps OTM prepayments can include cashout refinancings. In addition to defaults and curtailments, we also need to remove this contribution in order to isolate turnover-related prepayments.

Problem

- There is no obvious way to estimate OTM cashout refinancing rates from MBS prepayment data.

Solution

- Based on an analysis that compares Turnover rates implied by Existing Home Sales to OTM MBS prepayment rates, it appears that OTM cashout refis can contribute between 2-4% CPR.
- We specify a very simple model to capture the variation in these refi rates.

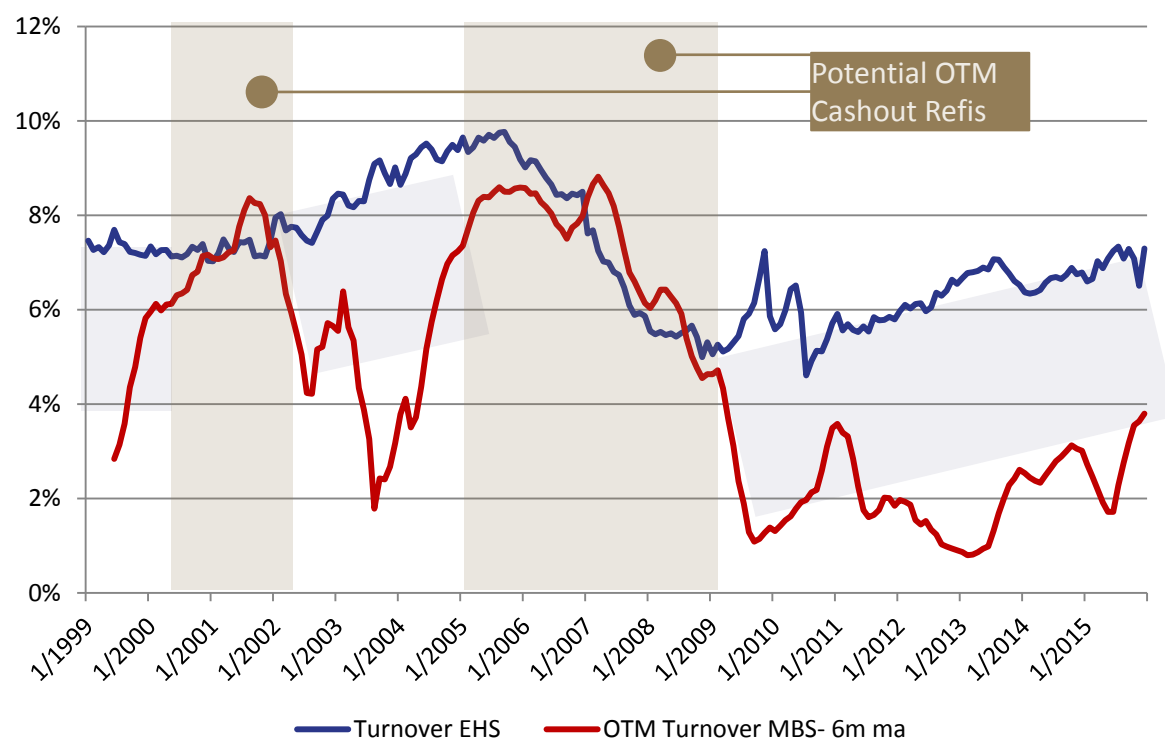
- Turnover EHS**

Existing Home Sales / Owner-occupied housing Stock

Turnover across all home owners, with or without mortgages (also, mortgage may be of any type)

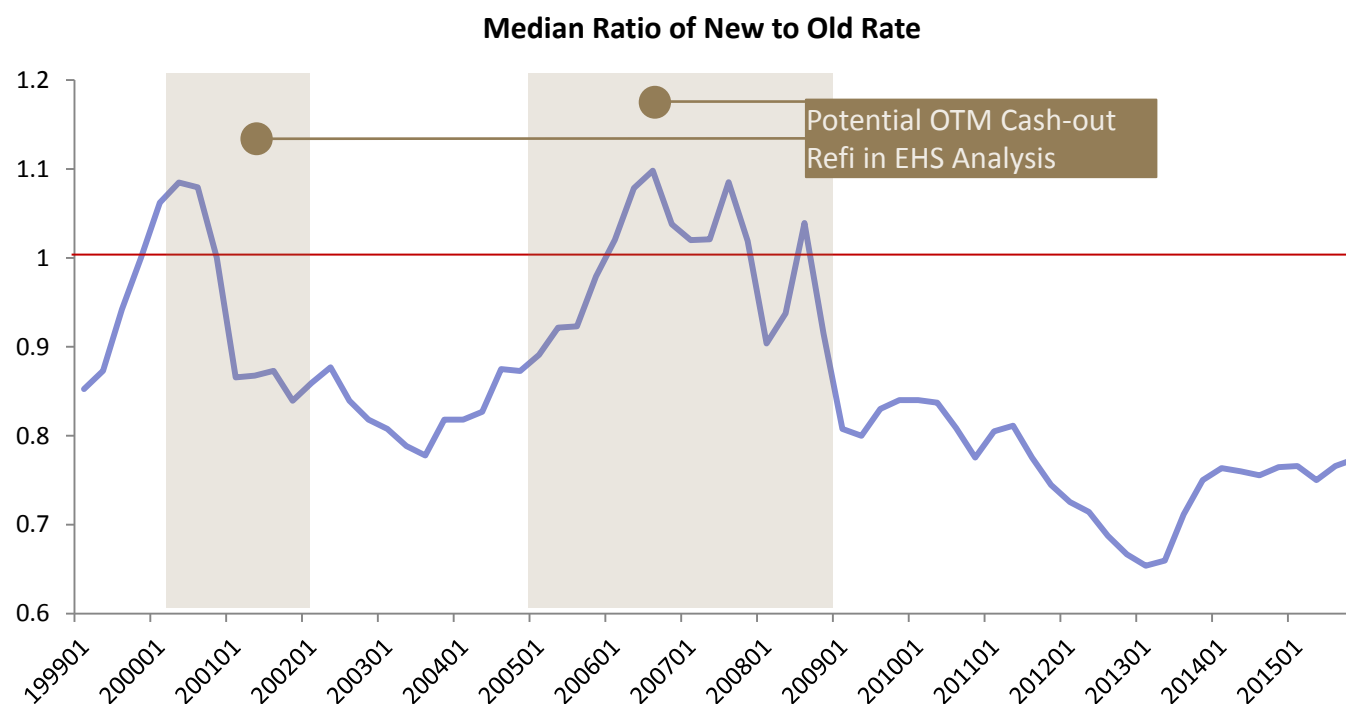
- Turnover MBS**

Prepayments on all pools at least 30 bps OTM after subtracting defaults and curtailments



- There is roughly a 4% difference between those two curves on average
- When the difference of two curves gets smaller, we potentially have an OTM Cashout refinance wave.

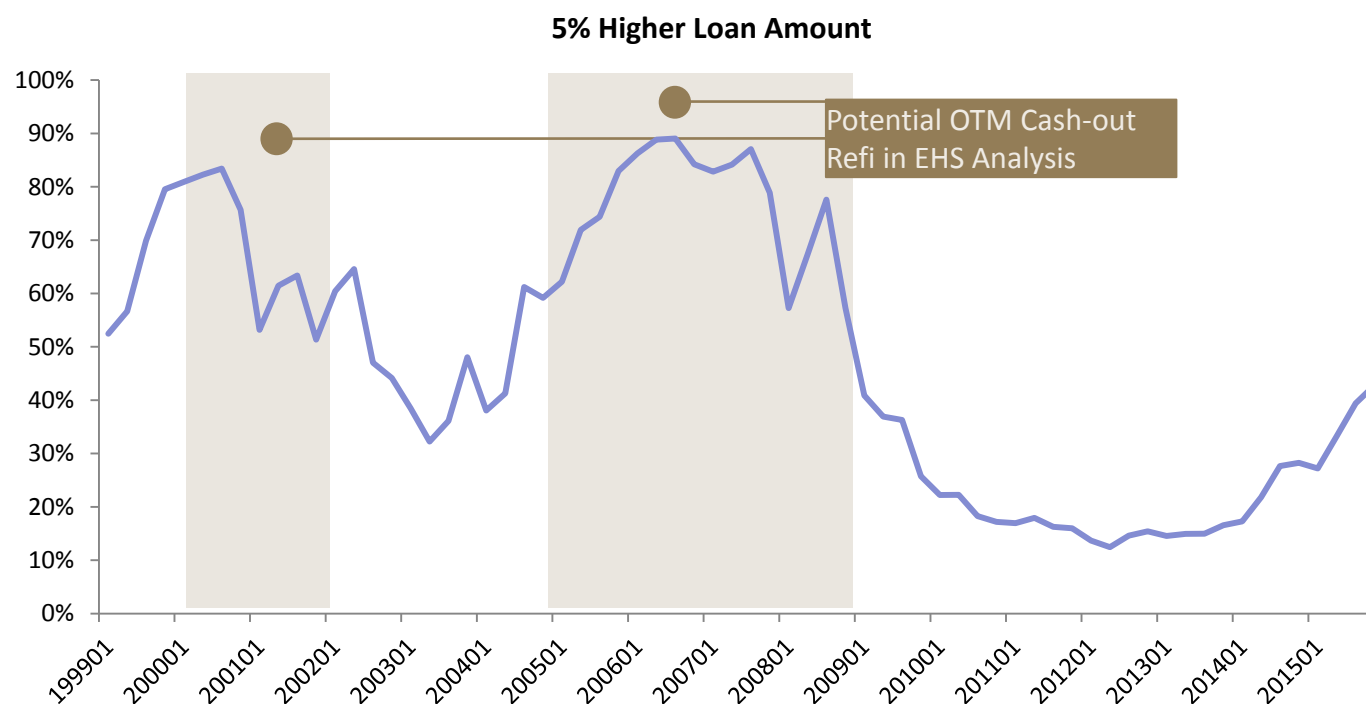
- **Ratio of New mortgage rate to Old rate**
 - Ratio of New to Old rate refers to the ratio of the interest rate of the new loan to the interest rate of the refinanced loan. Refinanced loans with adjustable-rate products are excluded.
 - When the Ratio of New to Old rate goes above 1, the interest rate of the refinanced loan is higher than that of the old loan. This likely corresponds to a significant amount of OTM Cashout Refis.



- Compare with the EHS Turnover analysis we can see that the potential OTM cash out refi period in the EHS analysis significantly overlaps the period when the ratio of the new to old rate ≥ 1 .

- 5% Higher Loan Amount

- "Higher Loan Amount" refers to loan amounts that were at least 5 percent greater than the amortized unpaid principal balance (UPB) of the original loan.
- When the percent of borrowers with 5% Higher Loan Amounts goes up, there are more cash-out refinances.



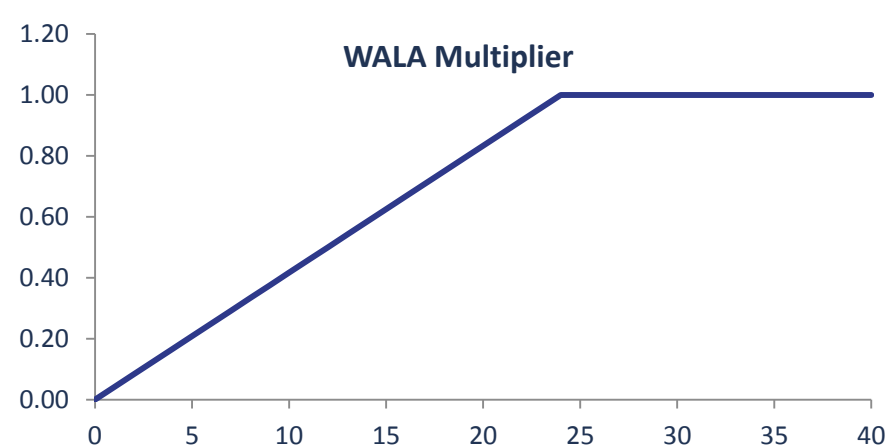
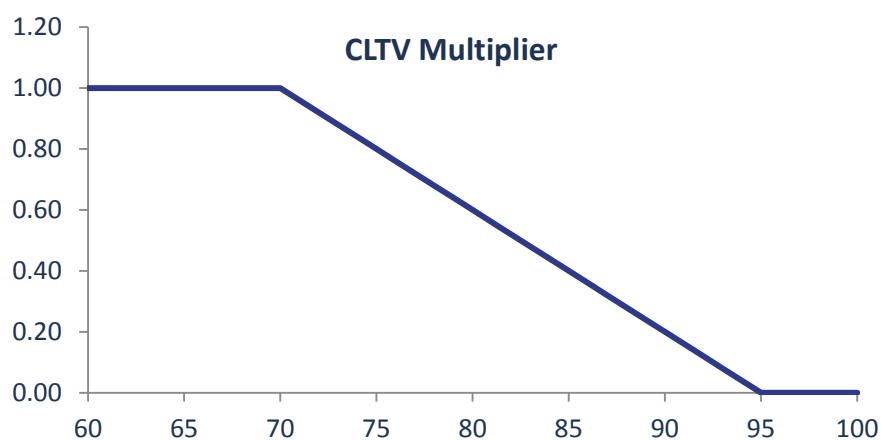
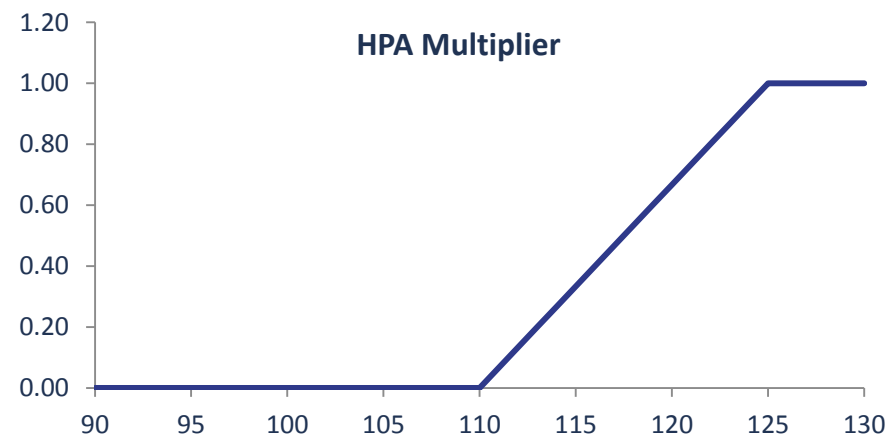
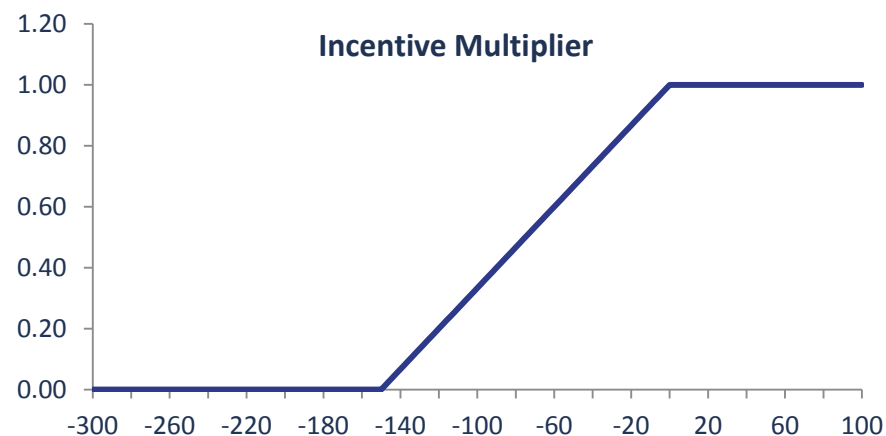
- Comparing with the EHS Turnover analysis we can see the potential OTM Cashout refi period in the EHS analysis significantly overlaps the period when the fraction of borrowers with 5% Higher Loan Amount goes up.

- Model to Estimate OTM Cashout Refis**

We introduced a four-factor model to estimate OTM Cashout Refis

OTM Cashout Refi = Incentive * HPA2yr * CLTV *WALA

Cashout refis are capped at 4% CPR.

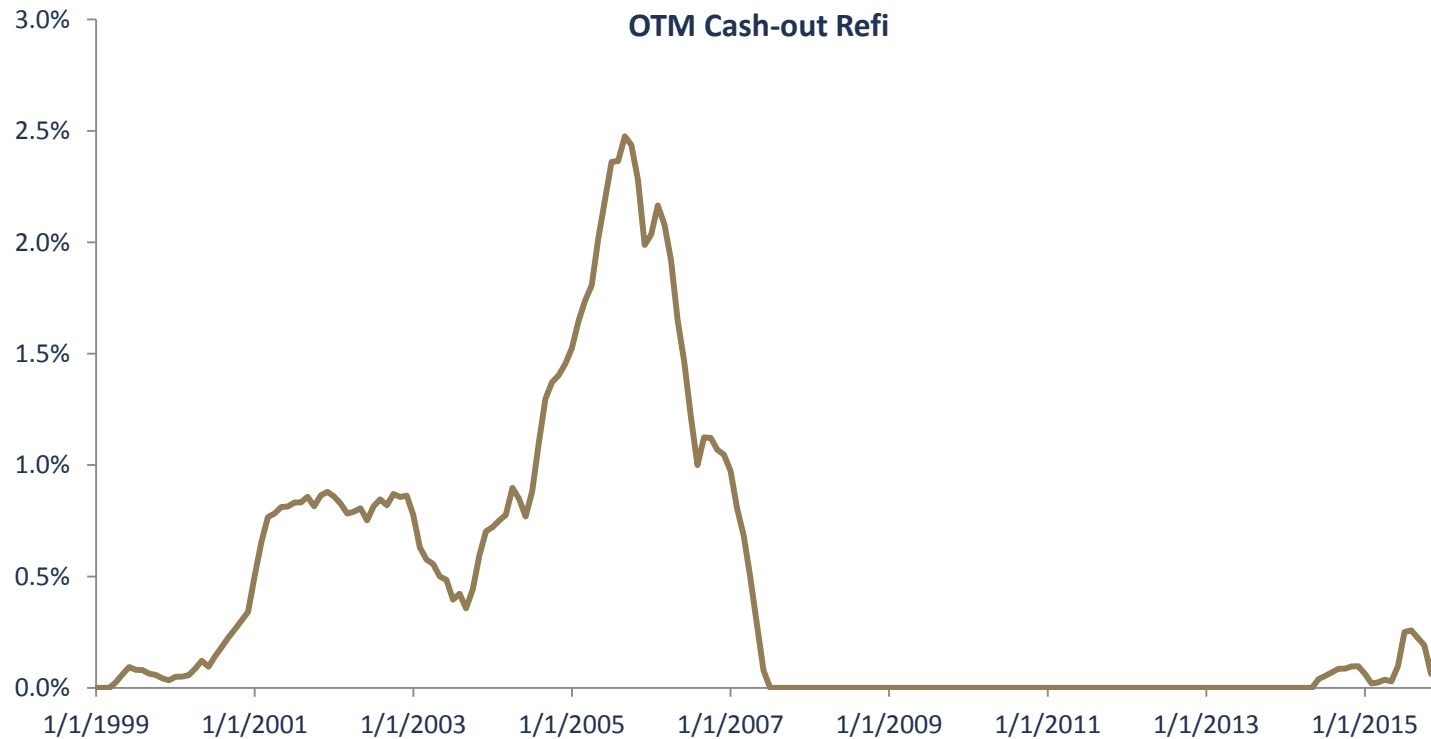


- Model to Estimate OTM Cash-out Refi**

We introduced a four-factor model to estimate OTM Cash-out Refi

$\text{OTM Cashout Refi} = \text{Incentive} * \text{HPA2yr} * \text{CLTV} * \text{WALA}$

Cashout refis are capped at 4% CPR



01 Seasonal Adj.

➤ Turnover shows clear seasonality. Removing the difference using seasonal adjustments helps in finding the spline functions for the model.

02 WALA

➤ The WALA ramp effect has turnover increasing as loan age goes up.

03 HPA2Yr

➤ In environments with greater house price appreciation, borrowers will have a higher chance of selling their houses.

04 Lock-in

➤ High levels of disincentive make it more costly for homeowners to move.

05 CLTV

➤ Borrower with higher CLTVs have more difficulty moving.

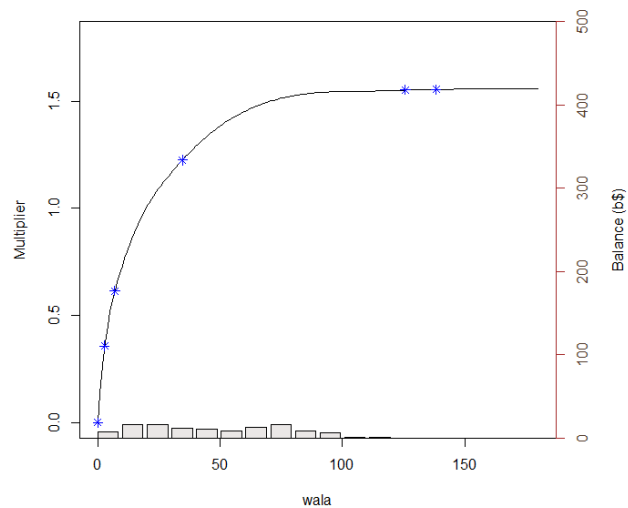
v2 of the turnover model includes the five prediction parameters listed on the left-hand side.

Sequentially fitting model with the five parameters, we can create a cubic spline curve for each one of these inputs.

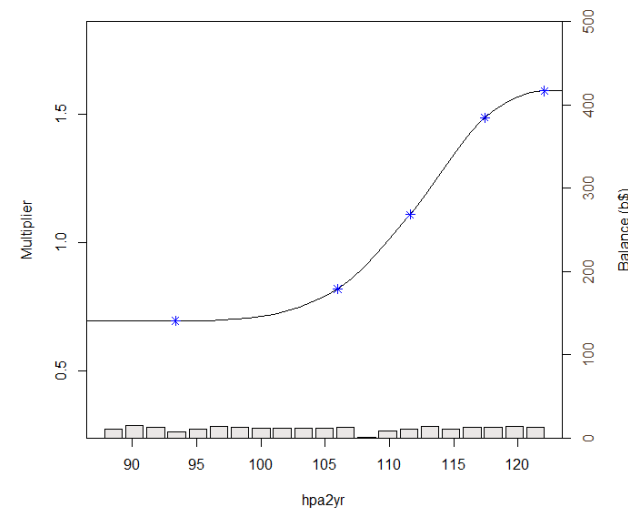
The parameter ordering is based on a single parameter regression that calculates R-square values.

Parameters that appear earlier in the order will have a more significant impact on our model output.

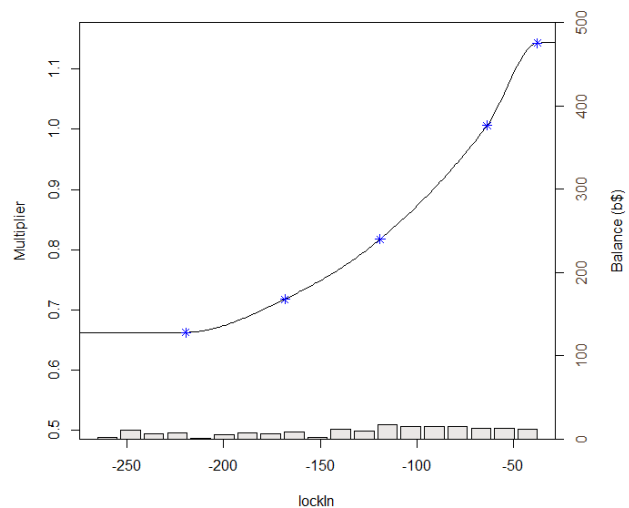
Spline Multiplier: wala



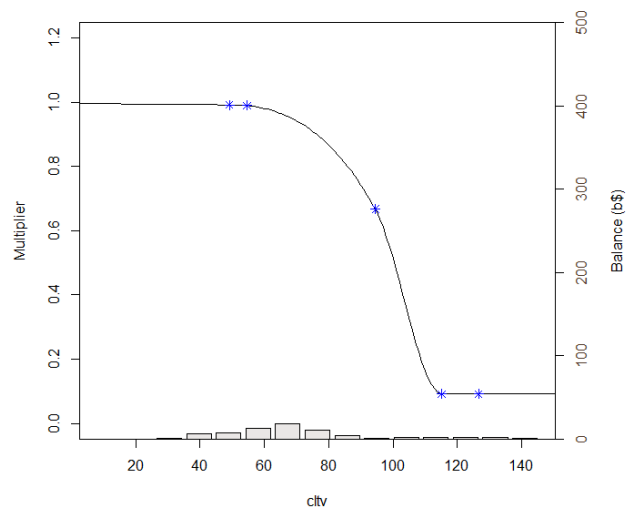
Spline Multiplier: hpa2yr



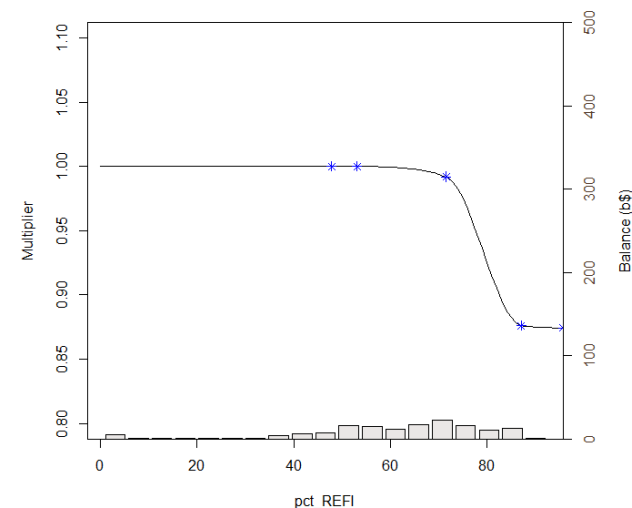
Spline Multiplier: lockln



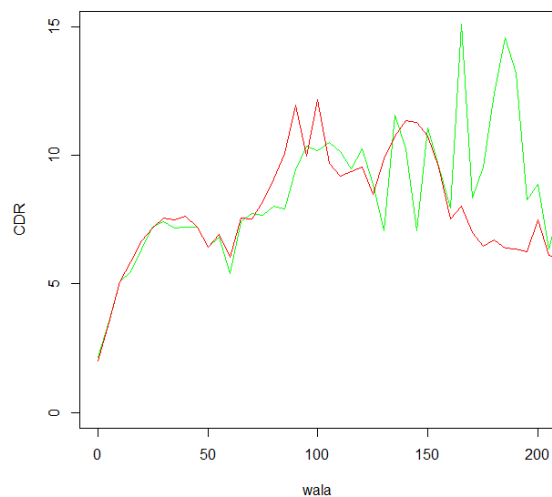
Spline Multiplier: cltv



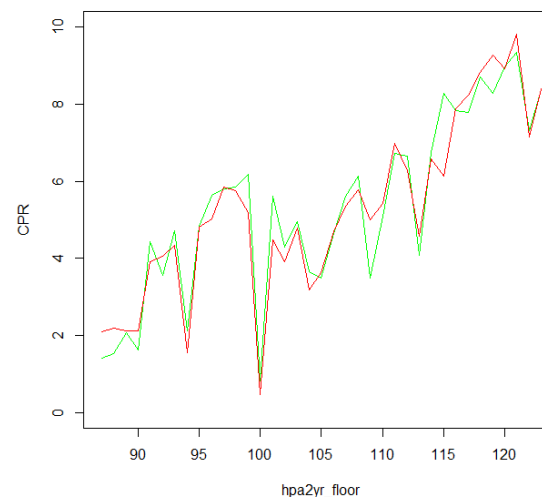
Spline Multiplier: pct_REFI



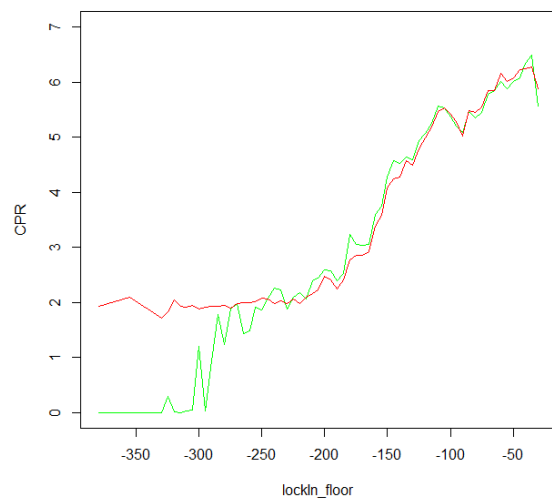
Tracking: turnover_pool_v1



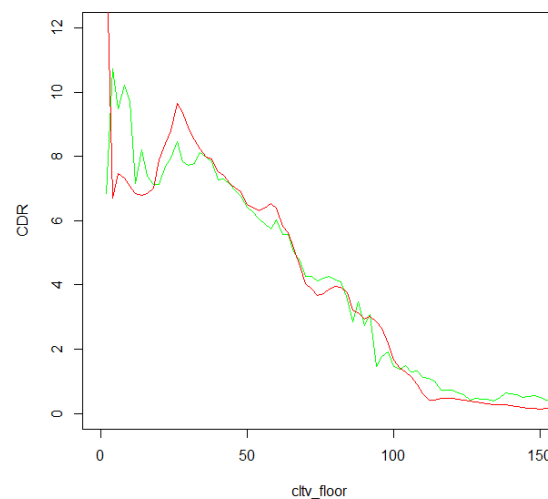
Tracking: turnover_pool_v2



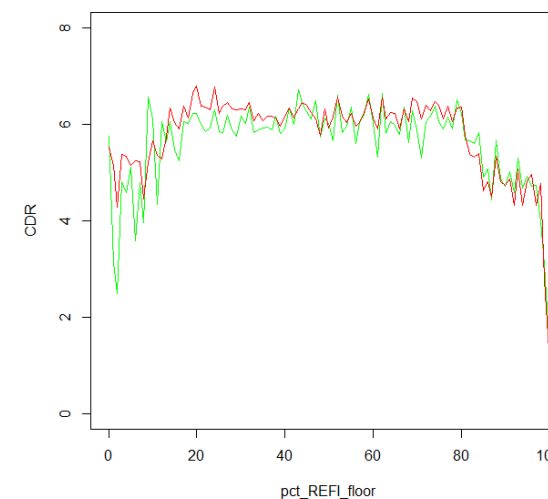
Tracking: turnover_pool_v2



Tracking: turnover_pool_v1

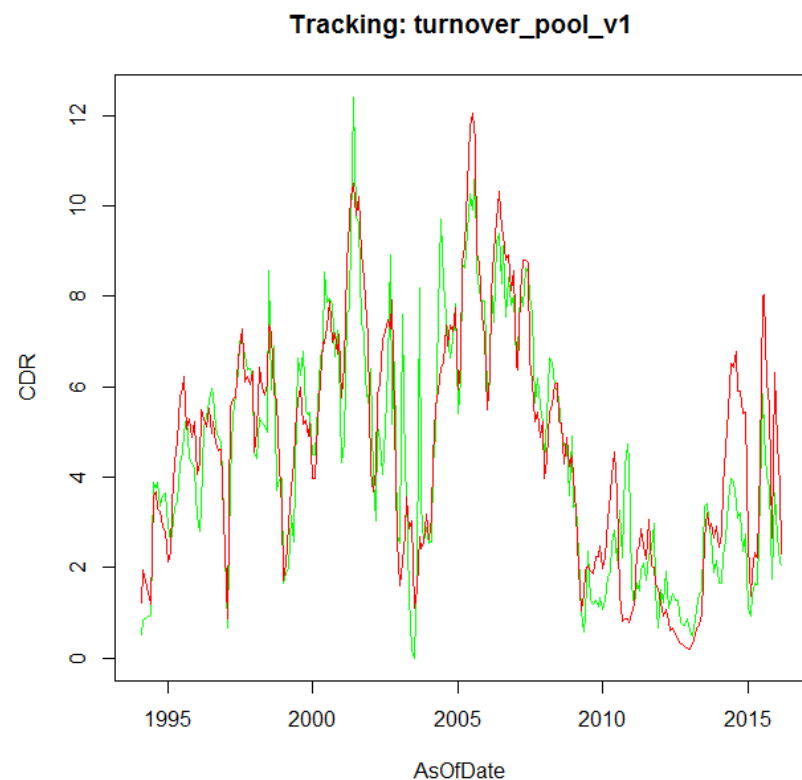


Tracking: turnover_pool_v2



Actual
Model

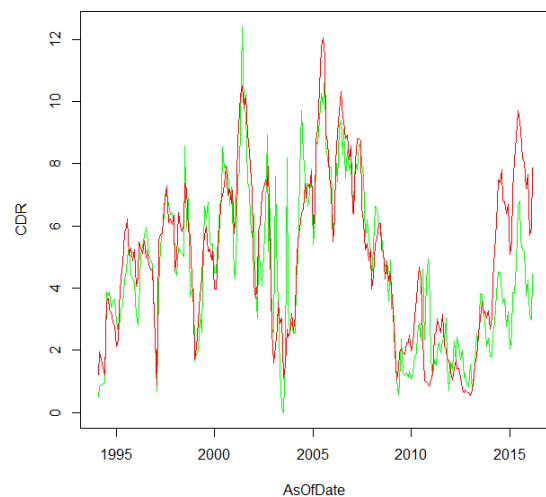
Model Tracking Performance – Overall (Conventional + Jumbo + CQ/CR)



Actual
Model

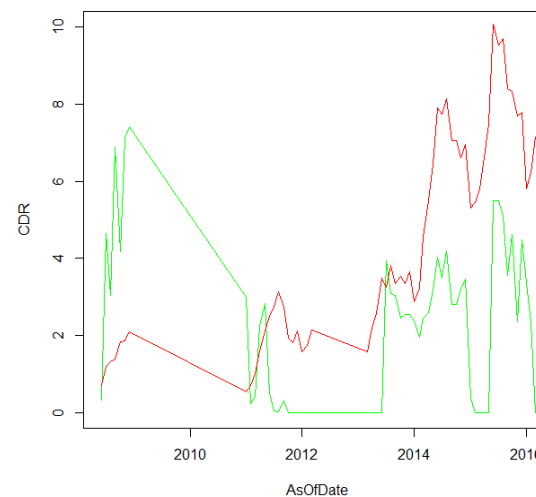
Model Tracking Performance

Conventional



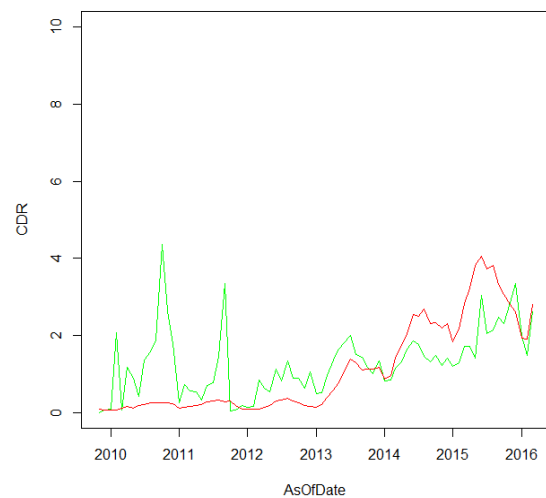
Conventional pool tracking shows model is 2% CPR faster in most recent one year.

Jumbo



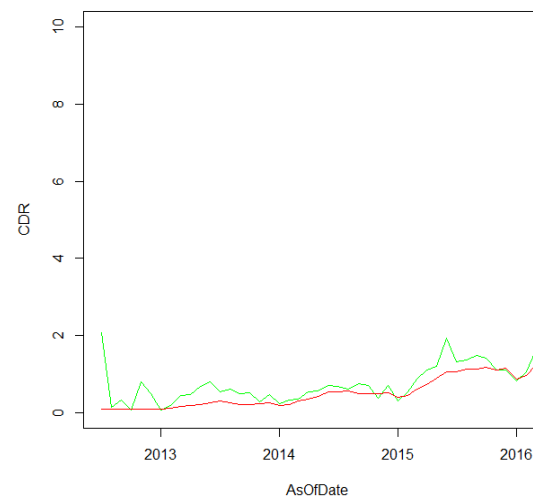
Few Jumbo pools with OTM incentive.

CQ



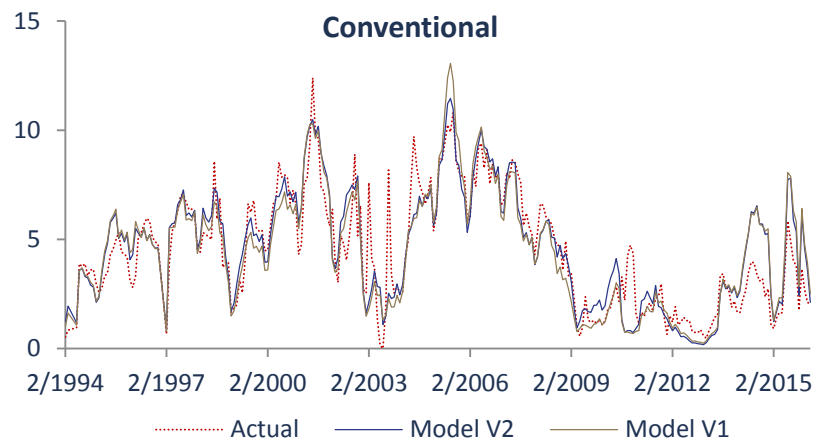
CQ pool tracking is more reliable now, as we fixed the incentive calculation for HARPed loans.

CR

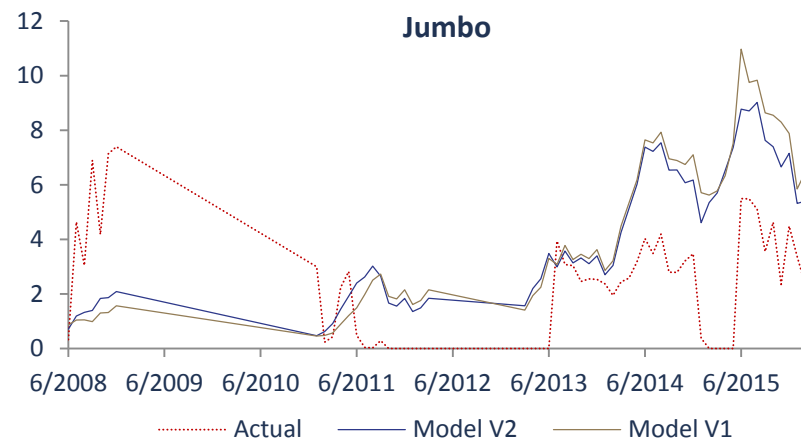


Similar to CQs

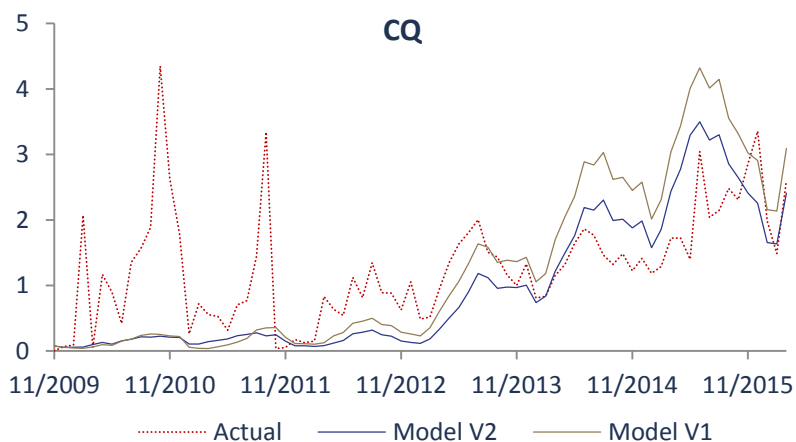
Actual
Model



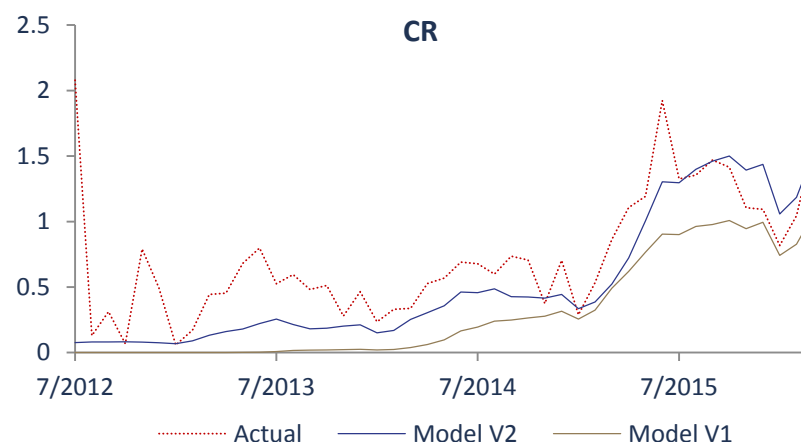
Model v1 and v2 results are quite similar (both are about 2% CPR too fast for recent time periods). v2 has a lower RMSE (1.218) versus v1 (1.303).



V2 is better



V2 is better

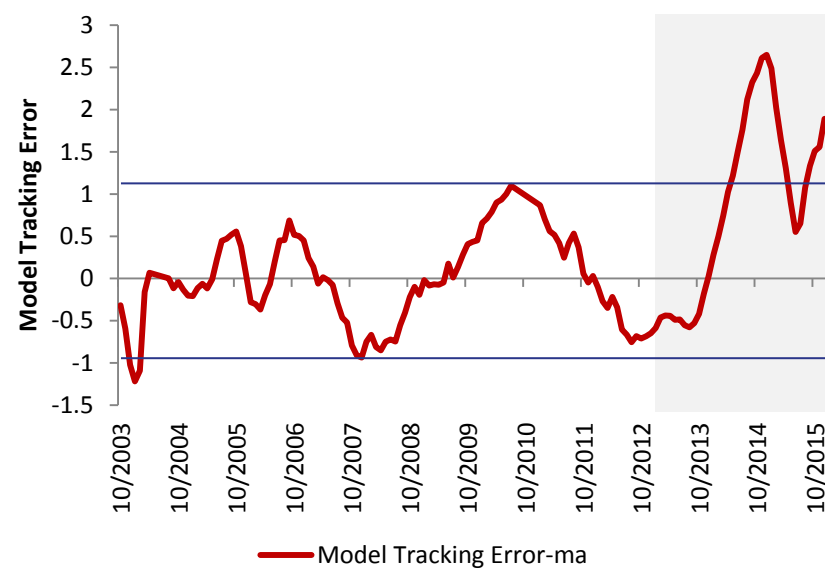
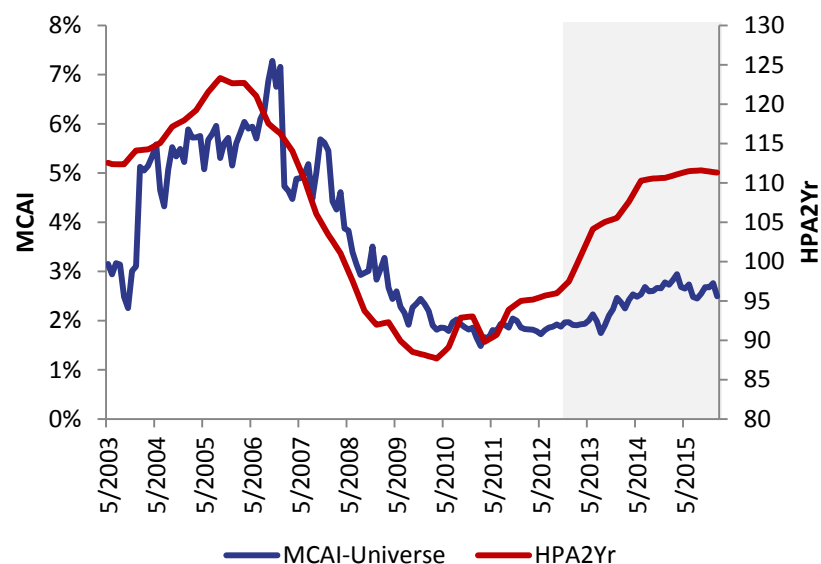


V2 is better

* All results based on data with updated incentive and OTM cash-out

	Min	Max
baselineConstant	5.823	
wala	0	1.570958
hpa	0.694794	1.530962
lockin	0.474579	1.148333
cltv	0.079905	0.987166
seasonality	0.683492	1.241067
Product of Multipliers	0	3.383621
CPR	0	19.71

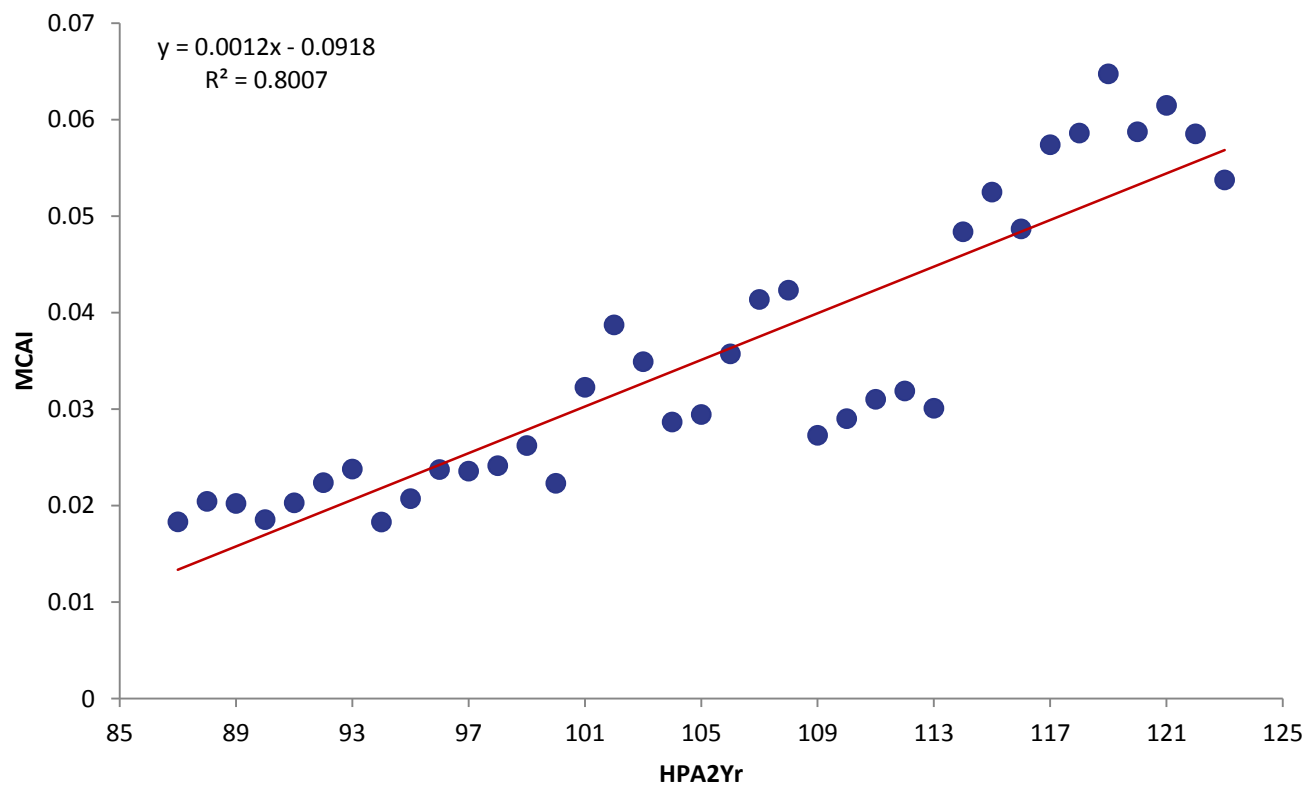
The Relationship between MCAI and HPA2Yr



In our current model, we use a two-year HPA moving average as a measure of mortgage credit availability. However, comparing the Mortgage Credit Availability Index and HPA2Yr curves, we can see a dispersion between them after 2013. At the same time, our model tracking error goes up.

We should look at changing the credit availability measure in a future version of model. MCAI would be one possible choice. In that case, since the MCAI is calculated by using origination information on all kinds of loan products(including non-agency products), we need to actively maintain the market share of different products. We will also need to smooth the MCAI so that a blip in production characteristics for a month doesn't have a profound impact in forward projections.

Predicting MCAI from HPA2Yr



As one would expect, there's a strong relationship between HPA2Yr and the MCAI. We can use this relationship to backfill MCAI data prior 2003 and to also forecast MCAI on the forward interest rate paths.